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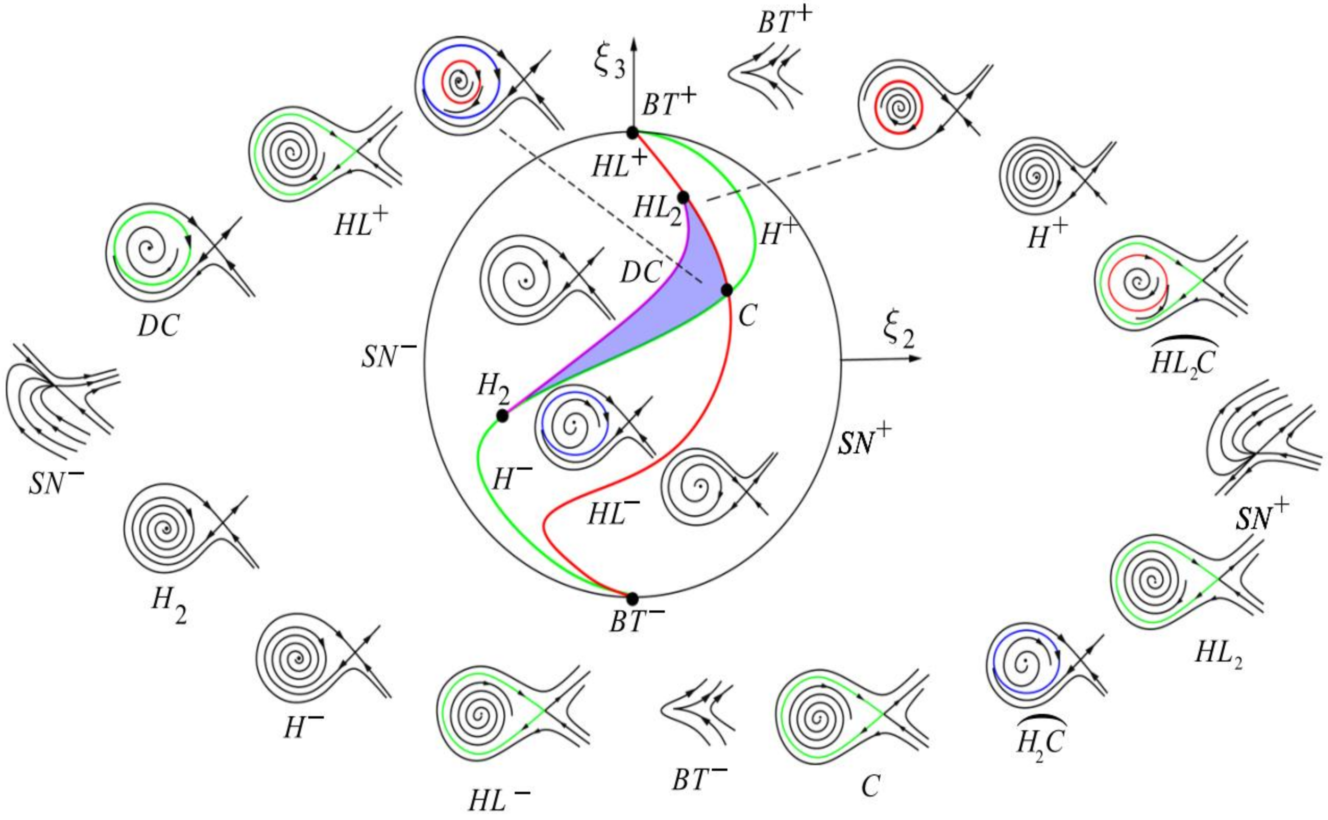
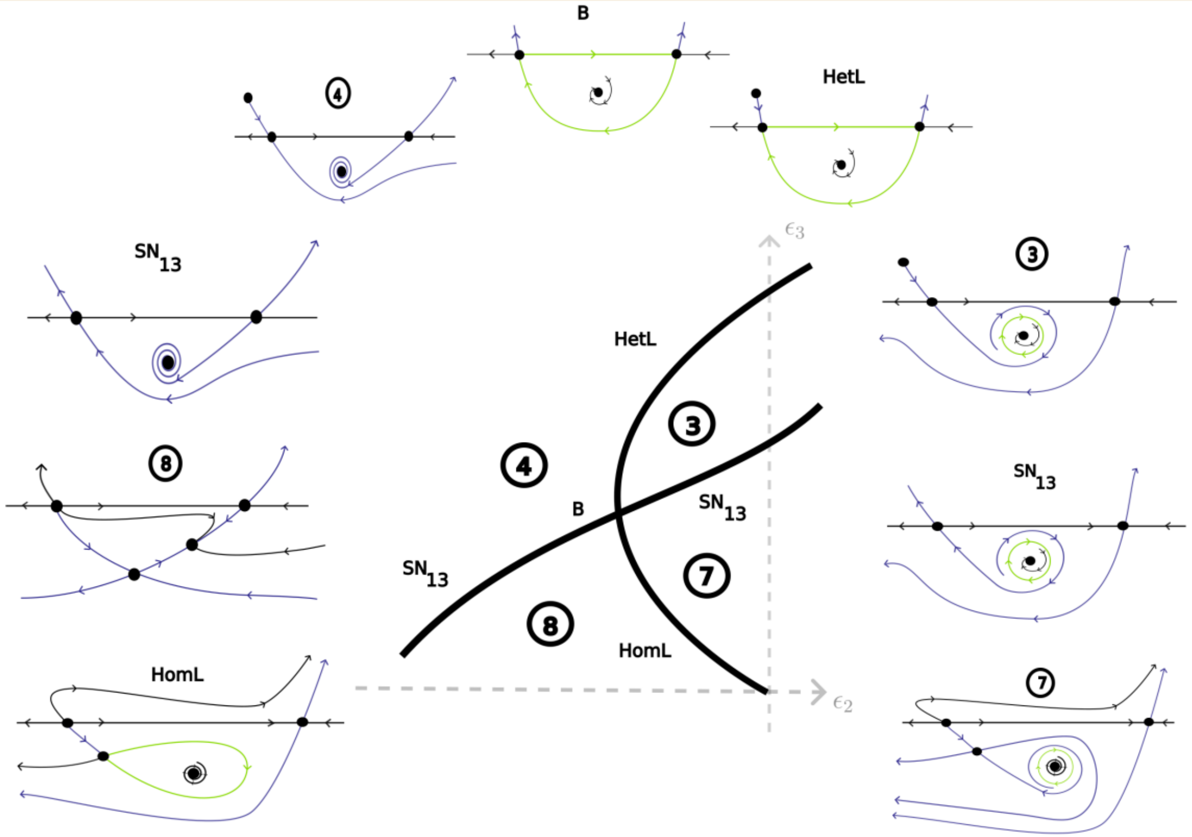
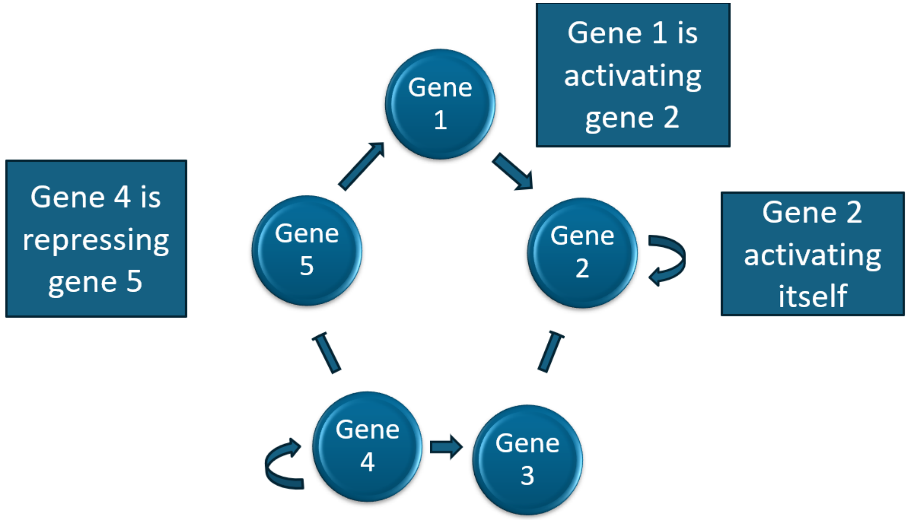
RESEARCH FOCUS:

MY RESEARCH LIES AT THE INTERSECTION OF DYNAMICAL SYSTEMS, DIFFERENTIAL EQUATIONS, AND MACHINE LEARNING WITH APPLICATIONS TO ECOLOGICAL AND BIOLOGICAL MODELING.





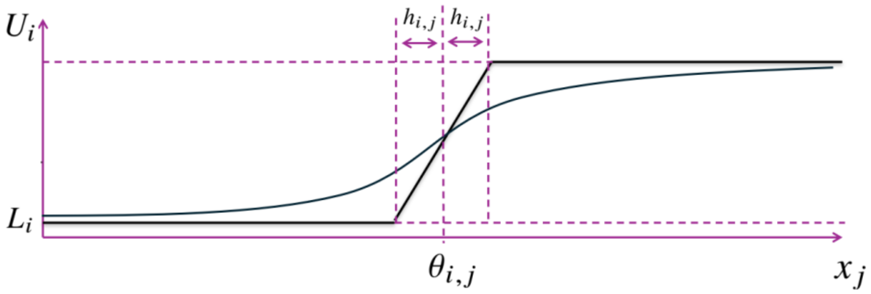


Let $x_i(t) \in \mathbb{R}_{\geq 0}$ denote the concentration of protein i at time t. The dynamics are governed by

$$\frac{dx_i}{dt} = -\gamma_i x_i + \sum_{j=1}^{m} s_{ij} H(\theta_{ij} - x_j)$$

where γ_i is the degradation rate, s_{ij} encodes the sign and strength of regulation from gene j to gene i, θ_{ij} are threshold concentrations, and $H(\cdot)$ is the Heaviside function:

$$H(z) = \begin{cases} 1, & z > 0, \\ 0, & z \leq 0. \end{cases}$$



The Heaviside function can be approximated by 2 smoother/less steep functions:

Hill function

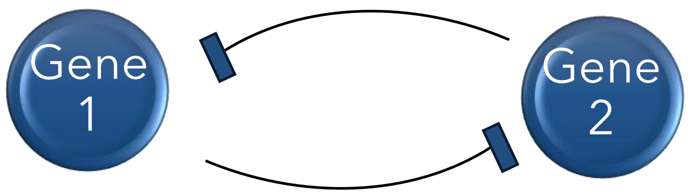
$$f_i(x_j) = (U_i - L_i)\,\frac{\lambda \theta_j^{d}}{\theta_j^{d} + x_j^{n}}$$

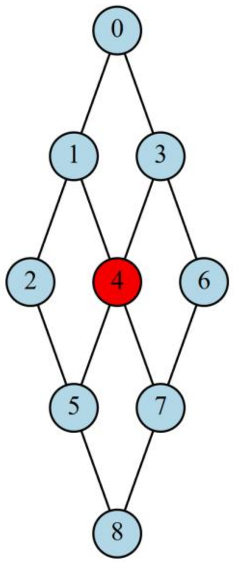
Ramp function

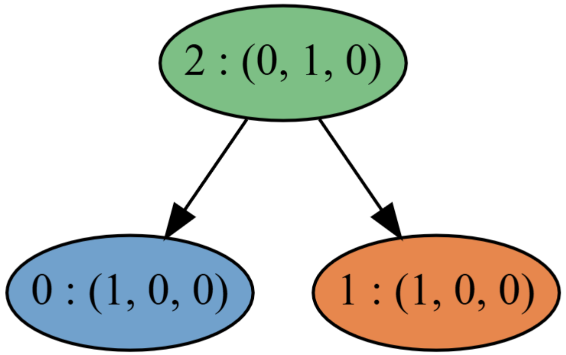
$$\begin{cases} L_i & x < \theta_{i,j} - h_{i,j} \\ L_{i,j}(x), & \theta_{i,j} - h_{i,j} \leq x \leq \theta_{i,j} + h_{i,j} \\ U_i & x > \theta_{i,j} + h_{i,j} \end{cases}$$

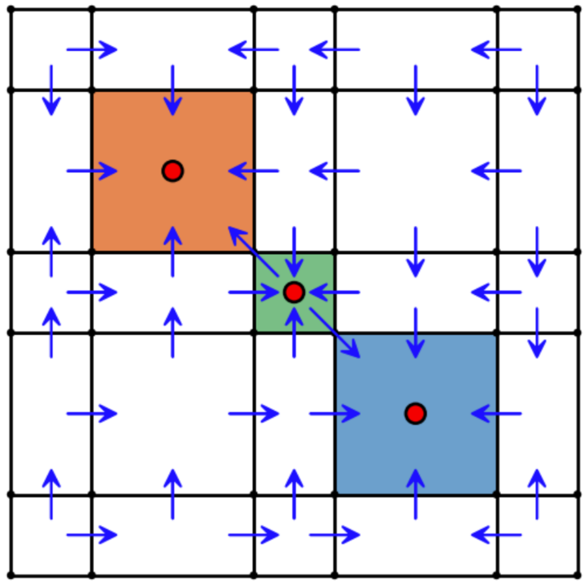
DSGRN:

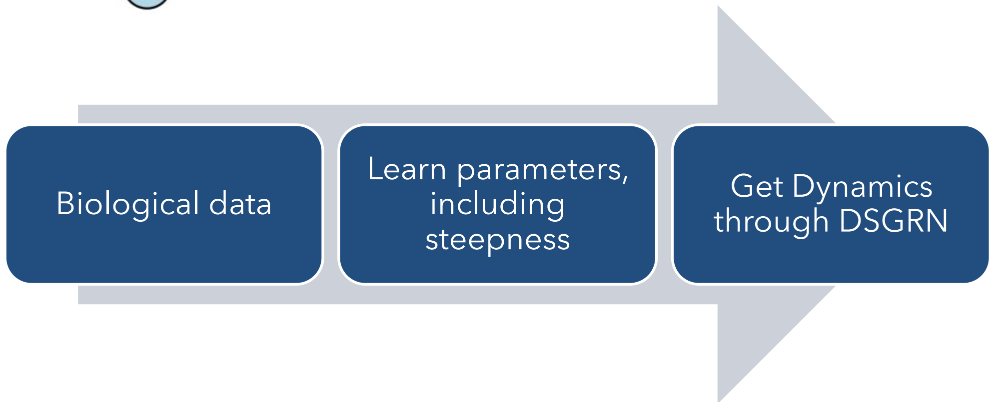
Dynamic Signatures Generated by Regulatory Networks (DSGRN) is a computational framework designed to analyze the global dynamics of gene regulatory networks without requiring precise parameter values. It partitions the high-dimensional parameter space into distinct regions and assigns each region a qualitative dynamical signature, such as a fixed point, a bistable region, or an oscillation, providing insight into the system's behavior under varying conditions.

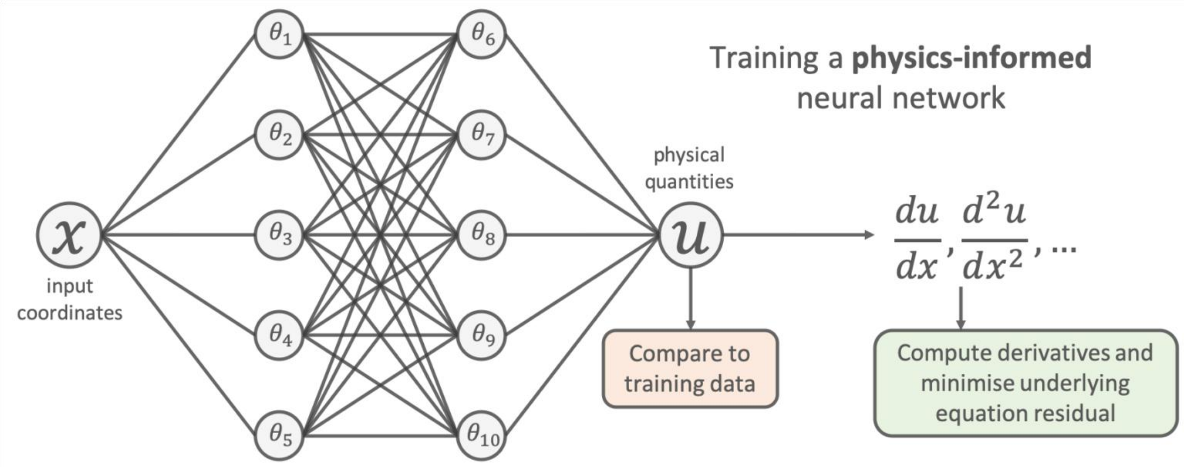


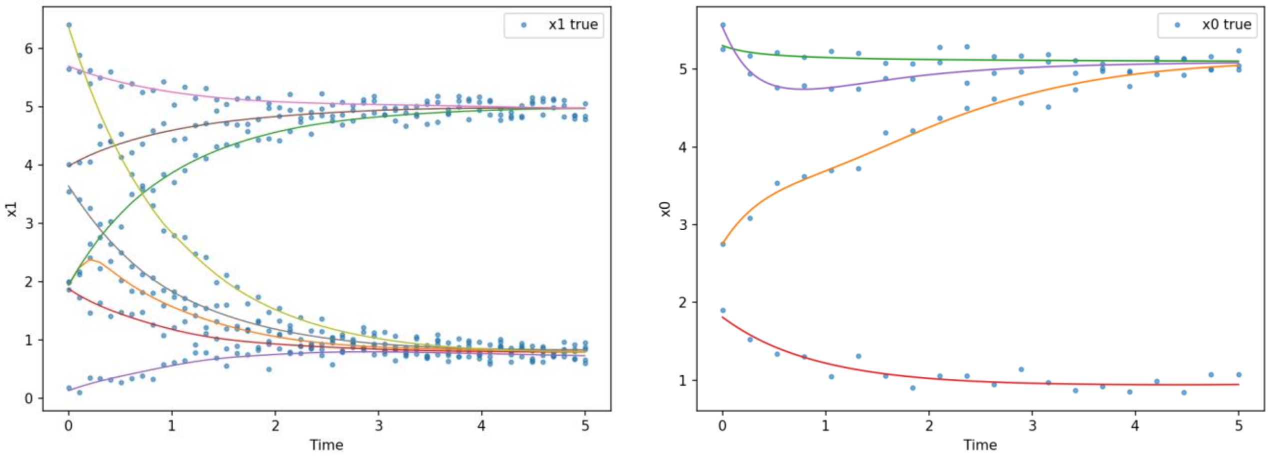




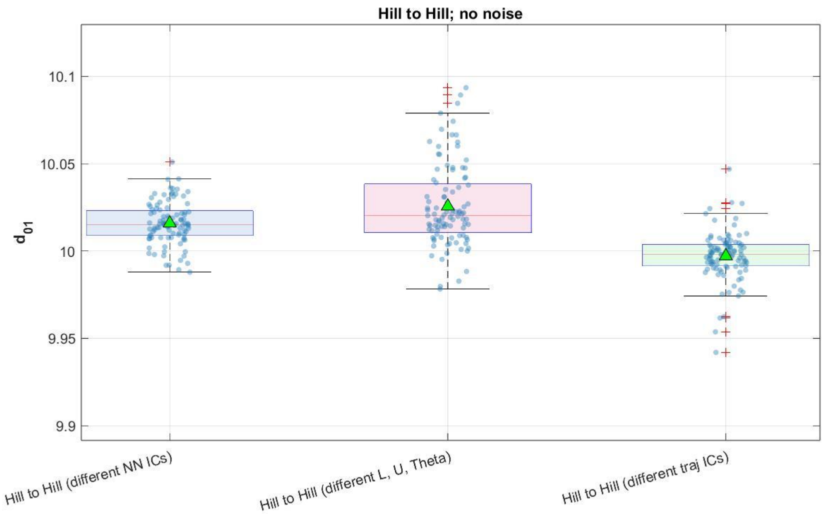




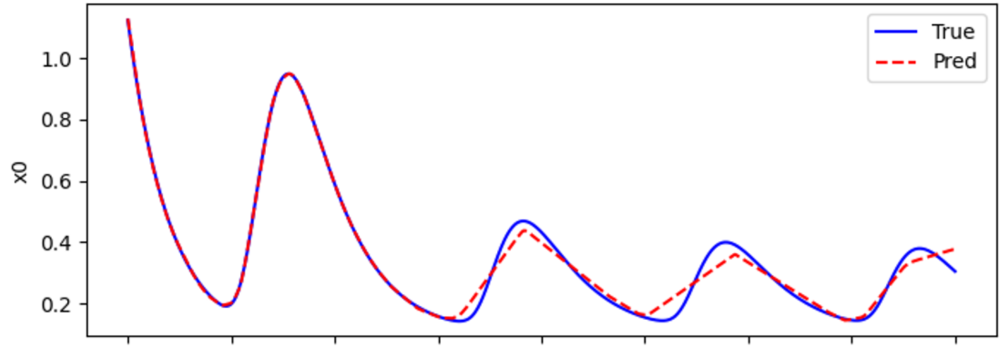




$$\mathcal{L} = \underbrace{\frac{1}{N_d}\sum_{i=1}^{N_d}\big(y_\theta(x_i) - y_i\big)^2}_{\text{data term}} + \underbrace{\lambda_f \frac{1}{N_c}\sum_{j=1}^{N_c}\big(\mathcal{N}[y_\theta](x_j)\big)^2}_{\text{physics residual}} + \underbrace{\lambda_b \frac{1}{N_b}\sum_{k=1}^{N_b}\big(\mathcal{B}[y_\theta](x_k)\big)^2}_{\text{BC/IC terms}},$$



No feature map



Fourier Feature map enabled